

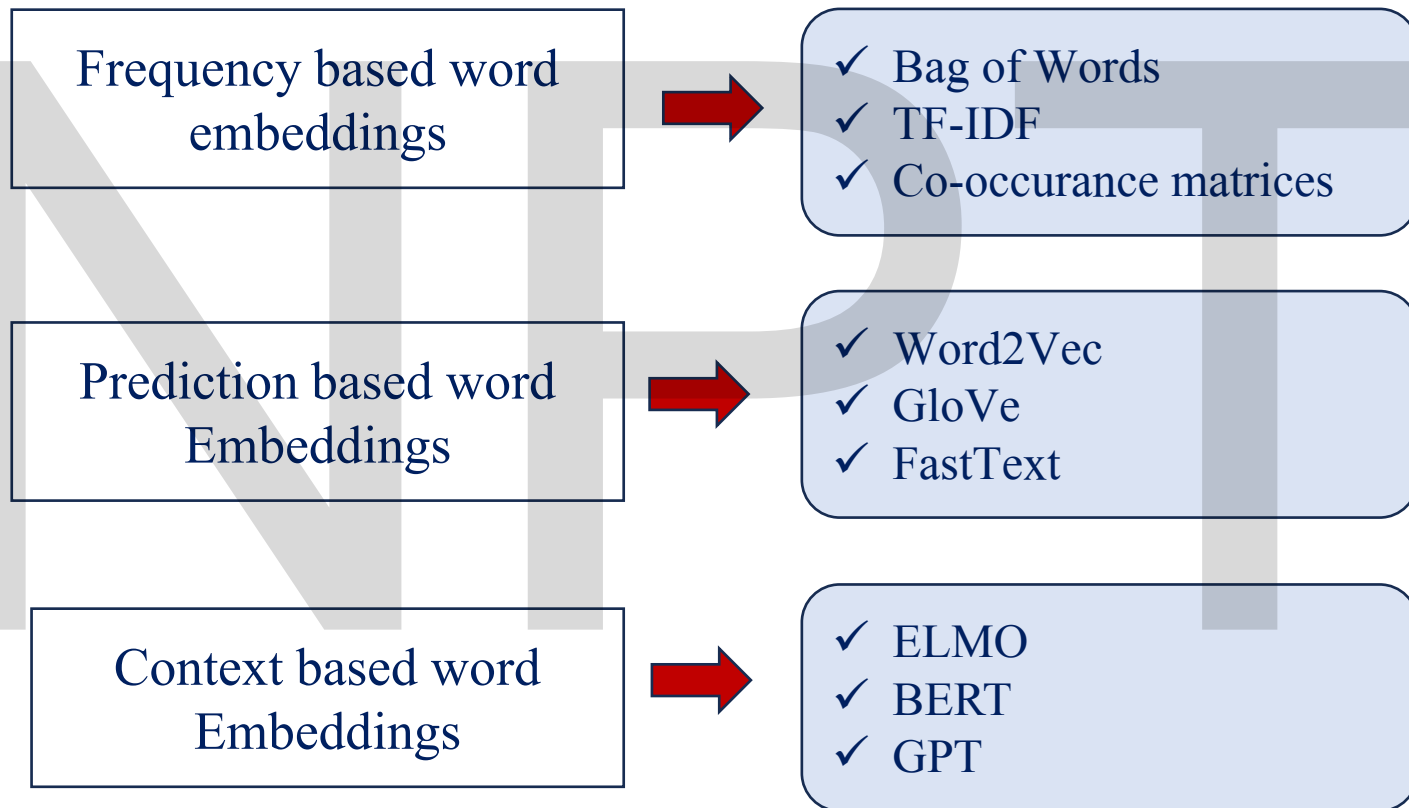
Word Embeddings in NLP

Contents

- ✓ GloVe Embedding
- ✓ FastText Embedding

NLP TREL

Types of Word Embeddings



GloVe Embedding

Key Idea

Learns word vectors from global word co-occurrence counts so that ratios of co-occurrence probabilities encode semantic relationships.

- ✓ Unsupervised algorithm for learning dense word embeddings
- ✓ Uses global co-occurrence patterns from a large text corpus.
- ✓ Captures semantic meaning and relationships between words

Working of GloVe Embedding

Text Corpus

“I like ice cream. I like chocolate ice cream.”

(This is the complete text from which word relationships are learnt)

Vocabulary

I, like, ice, cream, chocolate.

Context window

1 (look at 1 word left/right).

Co-occurrence Matrix X

Target words

Word\Context →	I	like	ice	cream	chocolate
I	0	2	0	0	0
like	2	0	1	0	1
ice	0	1	0	2	1
cream	0	0	2	0	0
chocolate	0	1	1	0	0

Computation of Co-occurrence Probabilities

Word\Context	I	like	ice	cream	chocolate	Row sum
I	0	2	0	0	0	2
like	2	0	1	0	1	2+1+1=4
ice	0	1	0	2	1	1+2+1=4
cream	0	0	2	0	0	2
chocolate	0	1	1	0	0	1+1=2

For a **target word** i , the probability of seeing context word j :



$$P(j | i) = \frac{X_{ij}}{\sum_k X_{ik}}$$

- $P(\text{like} | I) = 2/2 = 1$
- $P(\text{ice} | \text{like}) = 1/4 = 0.25$
- $P(\text{cream} | \text{ice}) = 2/4 = 0.5$
- $P(\text{ice} | \text{cream}) = 2/2 = 1$

These numbers show how strongly one word is associated with another word

Glove Algorithm

GloVe Equation

GloVe tries to find vectors w_i and w_j such that:

$$w_i^T w_j + b_i + b_j \approx \log(X_{ij})$$

frequent pairs $\rightarrow$ larger dot product,
rare pairs $\rightarrow$ smaller dot product

For target word “ice” and context word “cream”:

$$w_{ice}^T w_{cream} + b_{ice} + b_{cream} \approx \log(2)$$

For target word “ice” and context word “chocolate”:

$$w_{ice}^T w_{chocolate} + b_{ice} + b_{chocolate} \approx \log(1)$$

w_i : vector for target word i

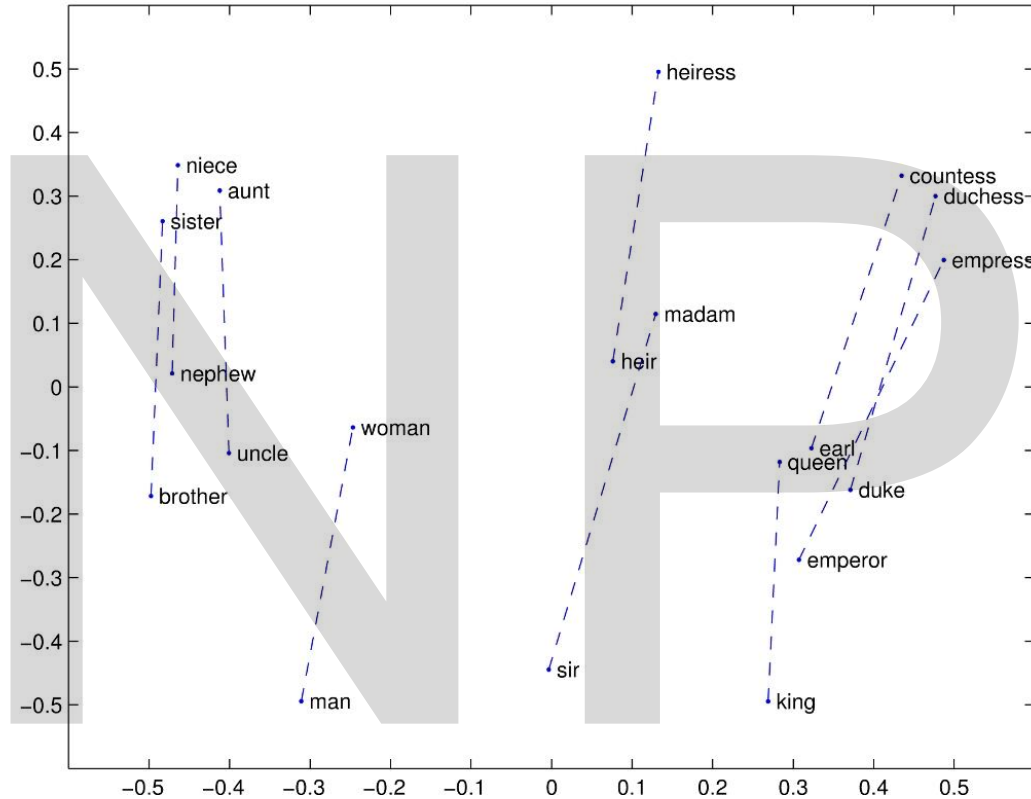
w_j : vector for context word j

b_i, b_j : bias terms

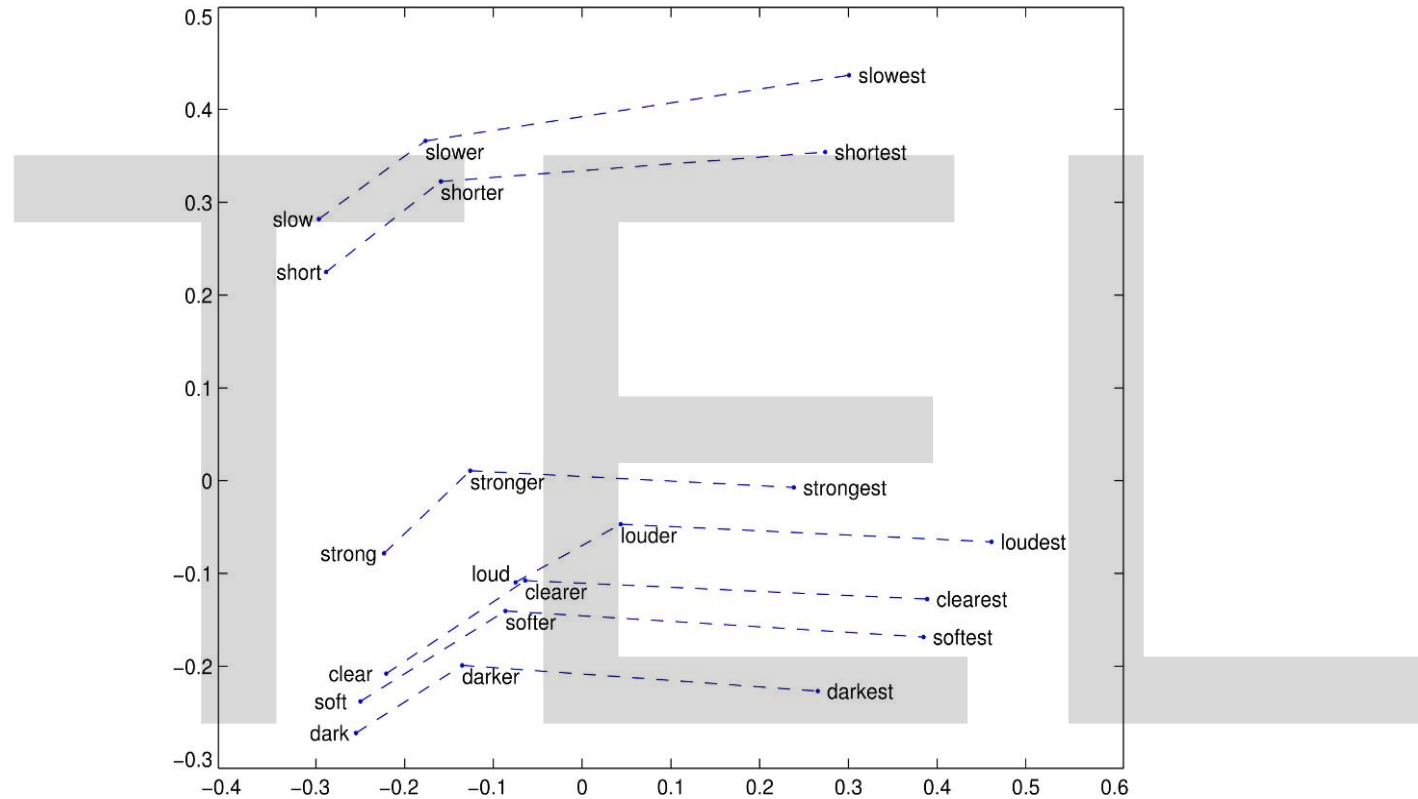
X_{ij} : Co-occurrence counts

Word\Context	I	like	ice	cream	chocolate
I	0	2	0	0	0
like	2	0	1	0	1
ice	0	1	0	2	1
cream	0	0	2	0	0
chocolate	0	1	1	0	0

Visualization of Glove Embedding Vectors



Man-Woman



Comparative-superlative

FastText Embedding

Developed by Facebook's AI Research Lab (FAIR).

Key Idea

Instead of assigning one vector per word, FastText represents a word as the sum of vectors of its character n-grams.

For example, the word "playing" with 3-gram (n=3) decomposition becomes
<pl, pla, lay, ayi, yin, ing, ng>

The word's embedding is then the sum (or average) of its n-gram embeddings + the word embedding itself.

Advantage:

- ✓ This makes it better at handling rare words and out-of-vocabulary (OOV) words.

Advantage of FastText Embedding

During training model saw:
“happy”, “unhappy”, “playing”

Never saw:
“unhappiness”

Word2Vec → OOV → no vector

FastText → split into n-grams:
<un, unh, hap, app, ppi, pin, ine, nes, ess, ss>



All these n-grams have embeddings → sum them
→ get an approximate but **usable vector** for “unhappiness”.

Handles typos
 (“plaing” instead of “playing”)

NPTEL

Summary

NLP TEL

*Next Session:
Hands on session on NLP*

GloVe Embedding

After training, the model outputs an embedding matrix where each word is represented by a dense vector. These vectors capture the semantic and syntactic relationships between words.

Word	Vector Embedding
I	[0.2,0.4,0.3]
Like	[0.1,0.6,0.7]
Ice	[0.9,0.4,0.5]
cream	[0.8,0.3,0.6]
chocolate	[0.6,0.2,0.9]

Numbers are representative